

Statistical Modeling and Air Quality Prediction Using Beijing Multi-Site Data

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Abstract

Air pollution is a major environmental and public health concern, especially in large urban areas where pollutant concentrations vary by location, season, time of day, and meteorological conditions. This project analyzes the Beijing Multi-Site Air Quality dataset to identify urban air pollution patterns and evaluate whether air pollutant and weather-related variables can be used to predict PM_{2.5} concentrations. The dataset contains hourly observations from 12 monitoring stations in Beijing from March 2013 through February 2017. PM_{2.5} was selected as the primary target variable because it is a widely used air-quality indicator and demonstrates meaningful variation across seasons and locations. The project follows an end-to-end statistical analysis workflow that includes data cleaning and preparation, exploratory data analysis, statistical hypothesis testing, model selection, model analysis, and recommendations. Exploratory analysis shows that PM_{2.5} is right-skewed, varies by station, is highest in winter and lowest in summer, and has strong positive correlations with PM₁₀, CO, NO₂, and SO₂. Statistical testing further indicates that PM_{2.5} differs significantly across seasons and has a significant negative relationship with wind speed. The model selection process compared linear regression, random forest, logistic regression, naive Bayes, and XGBoost approaches across regression and classification tasks. XGBoost was selected as the strongest overall model family, with the best all-station forecast regression model producing $R^2 = 0.515$, RMSE = 57.558, and MAE = 39.889.

Keywords: PM_{2.5}, air quality, Beijing, exploratory data analysis, statistical modeling, XGBoost

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Introduction

Air quality is an important environmental and public health issue. Fine particulate matter, commonly referred to as PM_{2.5}, is particularly important because small airborne particles can enter deep into the respiratory system and affect health. The U.S. Environmental Protection Agency (U.S. Environmental Protection Agency, 2026) explains that small particles can get deep into the lungs and, in some cases, enter the bloodstream. The World Health Organization (World Health Organization, n.d.) similarly describes air pollution as a major health concern because exposure affects the lungs, heart, brain, and other organs.

The purpose of this project is to perform an end-to-end statistical analysis of the Beijing Multi-Site Air Quality dataset. This dataset includes hourly measurements of six major air pollutants and six meteorological variables from multiple monitoring sites in Beijing (S. Chen, 2017). The project objective is to understand patterns in PM_{2.5} concentrations and develop a statistical or machine learning model to predict PM_{2.5} based on weather-related and pollutant variables.

This report is organized according to the required final project workflow. The data cleaning/preparation section describes how the raw station files were combined and treated for missing values. The exploratory data analysis section summarizes station-level, seasonal, time-series, and correlation patterns. The statistical analysis section evaluates whether observed seasonal differences and wind-speed relationships are statistically significant. The model selection and model analysis sections compare regression and classification approaches and explain why XGBoost was selected as the strongest model family. The conclusion and recommendations section connects the technical findings to environmental monitoring and decision-making.

Data Cleaning/Preparation

The Beijing Multi-Site Air Quality dataset was selected from the UCI Machine Learning Repository. The UCI documentation describes the dataset as a multivariate time-series dataset with 420,768 hourly observations, 12 monitoring stations, six major air pollutant variables, and six relevant meteorological variables. The data period spans March 1, 2013 through February 28,

2017, and missing data are denoted as NA (S. Chen, 2017).

The raw data were stored as 12 station-level CSV files. The first preparation step was to load the CSV files, extract the station name from each file, and concatenate the records into a single dataframe. This created a combined dataset with 420,768 observations and 18 original variables. A datetime feature was then created from the year, month, day, and hour fields. After adding the datetime column, the working dataframe contained 19 variables.

Table 1

Selected Dataset Variables

Variable group	Variables
Pollutants	PM2.5, PM10, SO2, NO2, CO, O3
Meteorological variables	TEMP, PRES, DEWP, RAIN, wd, WSPM
Time variables	year, month, day, hour, datetime
Location variable	station

Note. The dataset contains hourly air pollution and meteorological measurements across 12 Beijing monitoring stations.

Missing values were reviewed across all columns. Pollutant variables had the largest missing rates, with CO having the highest missing percentage. Weather variables had smaller missing percentages. Table 2 summarizes the primary missing value counts.

Table 2*Missing Value Summary Before Imputation*

Variable	Missing count	Missing %
CO	20,701	4.92
O3	13,277	3.16
NO2	12,116	2.88
SO2	9,021	2.14
PM2.5	8,739	2.08
PM10	6,449	1.53
wd	1,822	0.43
WSPM	318	0.08

Note. Only the most relevant missing variables are shown.

Numerical missing values were imputed using station-level medians. This choice was appropriate because pollutant variables are right-skewed and contain extreme values, so the median is less sensitive to unusually high pollution episodes than the mean. The categorical wind direction variable was imputed using the station-level mode. After imputation, the cleaned dataset had no remaining missing values and was saved as `beijing_air_quality_clean.csv` for additional analysis and modeling. Imputation here runs the risk of data leakage, but it was done for the sake of dataset consistency during model comparison.

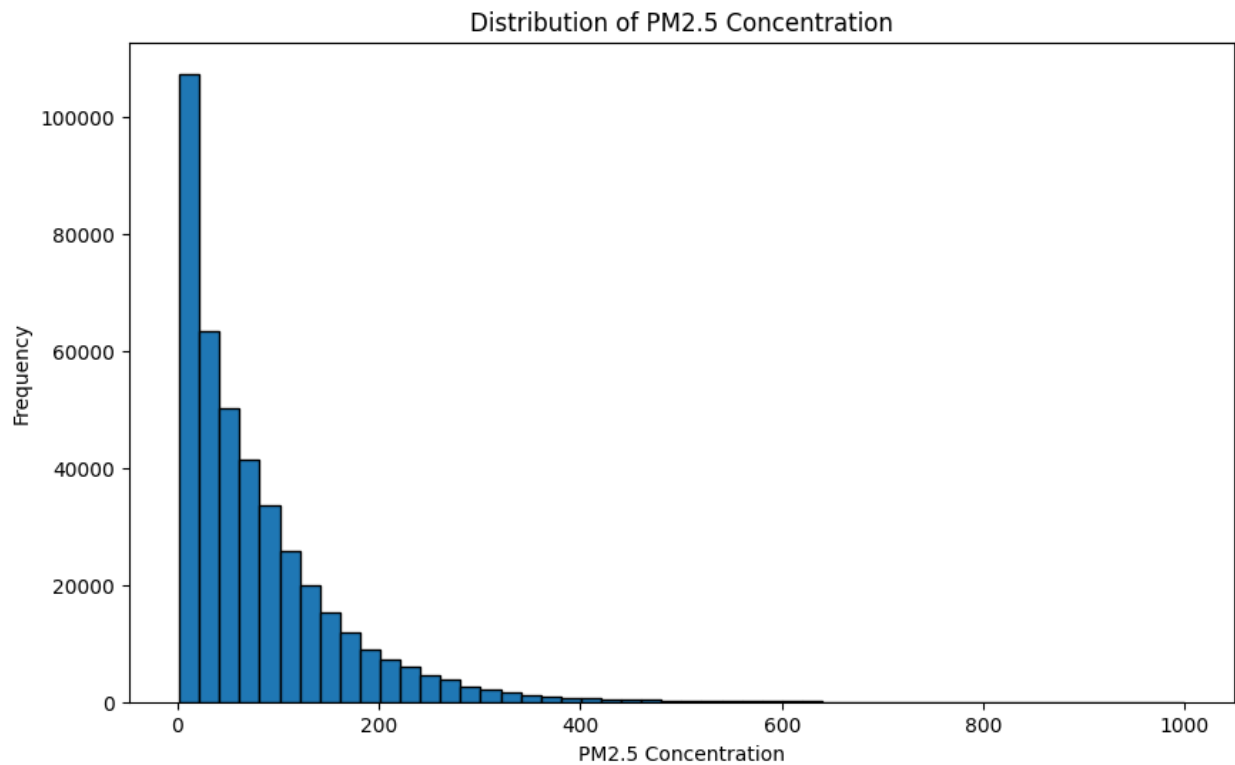
Additional feature engineering was also performed. A season variable was created from the month variable using the categories winter, spring, summer, and fall. This feature supports seasonal comparison and hypothesis testing. For model preparation, hour and month were also represented using sine and cosine transformations because time variables repeat in cycles; for example, hour 23 and hour 0 are adjacent even though their raw numeric values appear far apart.

Exploratory Data Analysis

Exploratory data analysis was conducted to understand PM2.5 patterns before model development. The EDA focused on PM2.5 distribution, station-level differences, seasonal patterns, time-series trends, wind-speed relationships, and correlations among pollutant and meteorological variables. This section connects directly to model selection by identifying variables that may help predict PM2.5.

PM2.5 Distribution

The PM2.5 distribution is strongly right-skewed. Most observations are concentrated at lower PM2.5 concentrations, while a smaller number of observations show very high pollution values. The notebook summary statistics show a mean PM2.5 concentration of approximately 79.79 and a median of 55.00. Since the mean is higher than the median, the distribution demonstrates positive skewness. This matters for modeling because extreme pollution events can influence error metrics and model fit.

Figure 1*Distribution of PM2.5 Concentration*

Note. Most observations occur at lower PM2.5 levels, with a long right tail indicating occasional extreme pollution events.

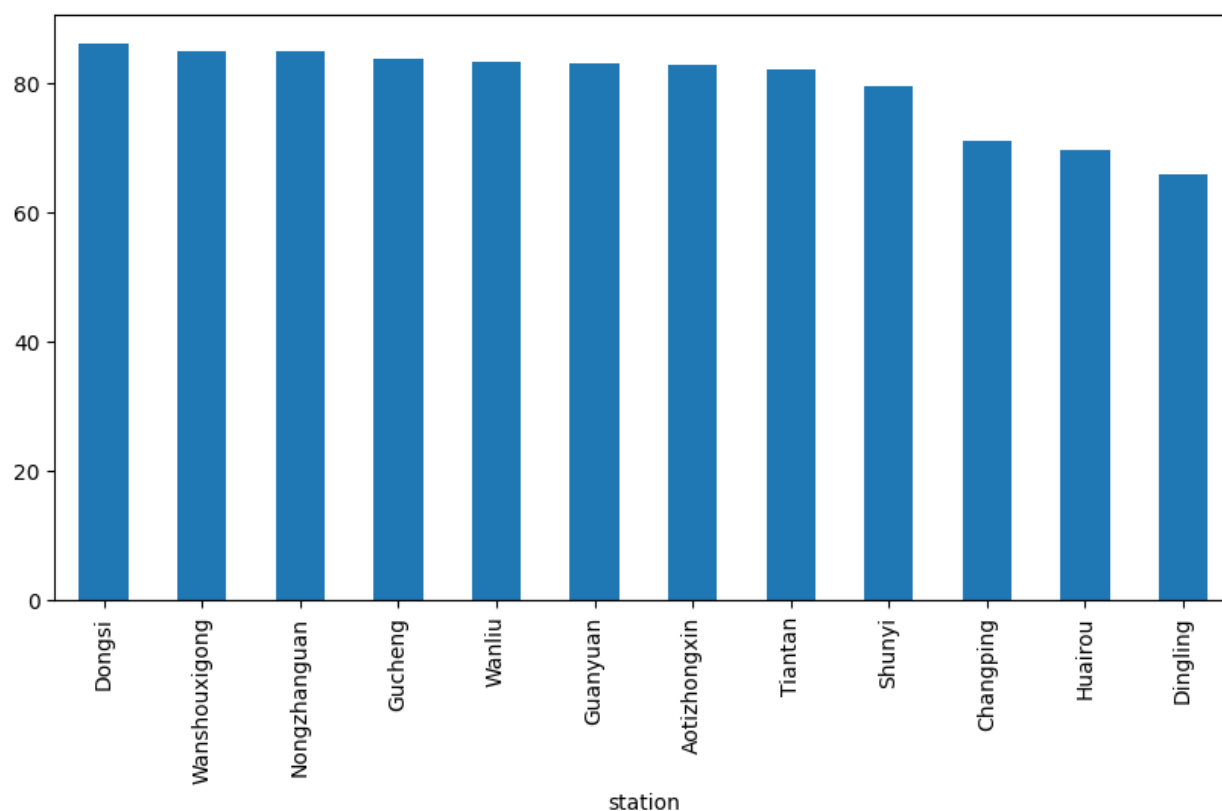
Station-Level PM2.5 Patterns

Average PM2.5 levels varied across the 12 monitoring stations. The highest mean PM2.5 values in the current notebook output were observed at Dongsì, Wanshouxigong, Nongzhanguan, and Gucheng. The lowest averages were observed at Dingling, Huairou, and Changping. These differences suggest that PM2.5 concentrations are not evenly distributed across Beijing monitoring locations. Station-level variation may reflect geographic conditions, traffic density, local emission sources, and weather patterns.

Table 3*PM2.5 Summary Statistics by Station*

Station	Mean	Median	Standard deviation
Dongsi	86.19	61.00	86.58
Wanshouxigong	85.02	60.00	85.98
Nongzhanguan	84.84	59.00	86.23
Gucheng	83.85	60.00	82.80
Dingling	65.99	41.00	72.27

Note. Table shows selected high and low station-level averages from the data preparation notebook.

Figure 2*Average PM2.5 Concentration by Monitoring Station*

Note. The station-level bar chart shows location-based variation in PM2.5 concentrations.

Seasonal PM2.5 Patterns

Seasonal analysis showed that PM2.5 concentrations were highest in winter and lowest in summer. The seasonal mean PM2.5 concentration was approximately 95.48 in winter, 82.33 in fall, 76.97 in spring, and 64.67 in summer. This pattern suggests that meteorological conditions and seasonal activity patterns may influence PM2.5 levels. Winter pollution may be related to lower atmospheric mixing, heating demand, and weather conditions that trap pollutants near the ground.

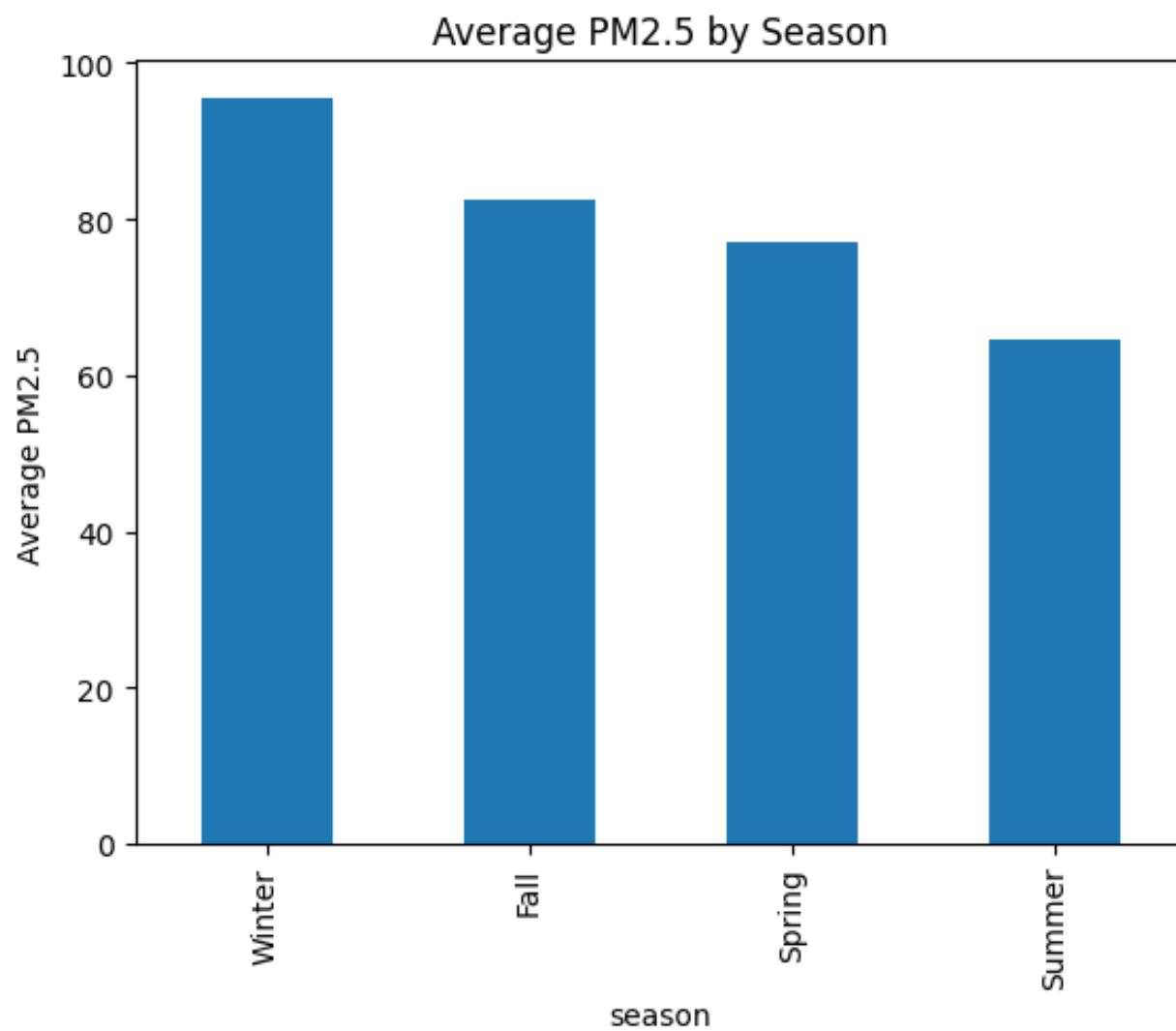
Table 4*PM2.5 Summary Statistics by Season*

Season	Count	Mean	Median	Standard deviation
Winter	101,907	95.48	56.00	107.66
Fall	102,746	82.33	56.00	81.44
Spring	103,705	76.97	58.00	68.83
Summer	103,671	64.67	51.00	52.65

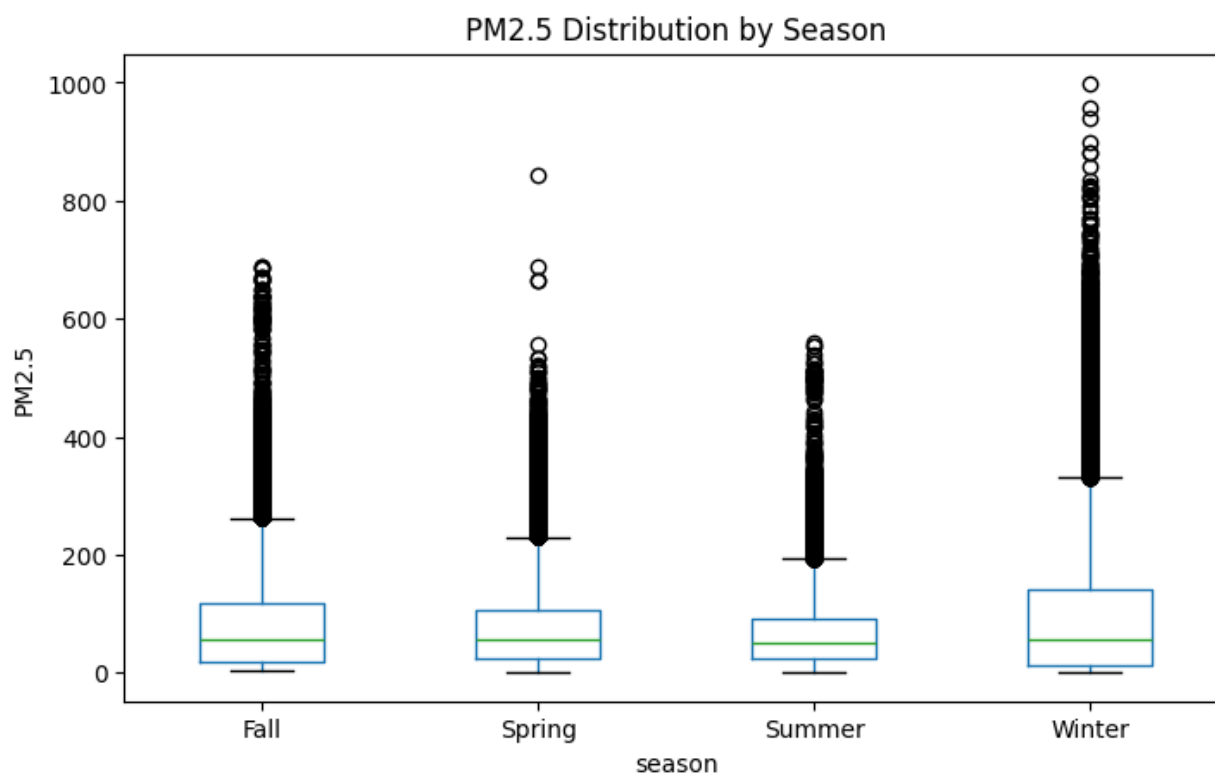
Note. Season was derived from month: winter = December–February, spring = March–May, summer = June–August, and fall = September–November.

Figure 3

Average PM2.5 by Season



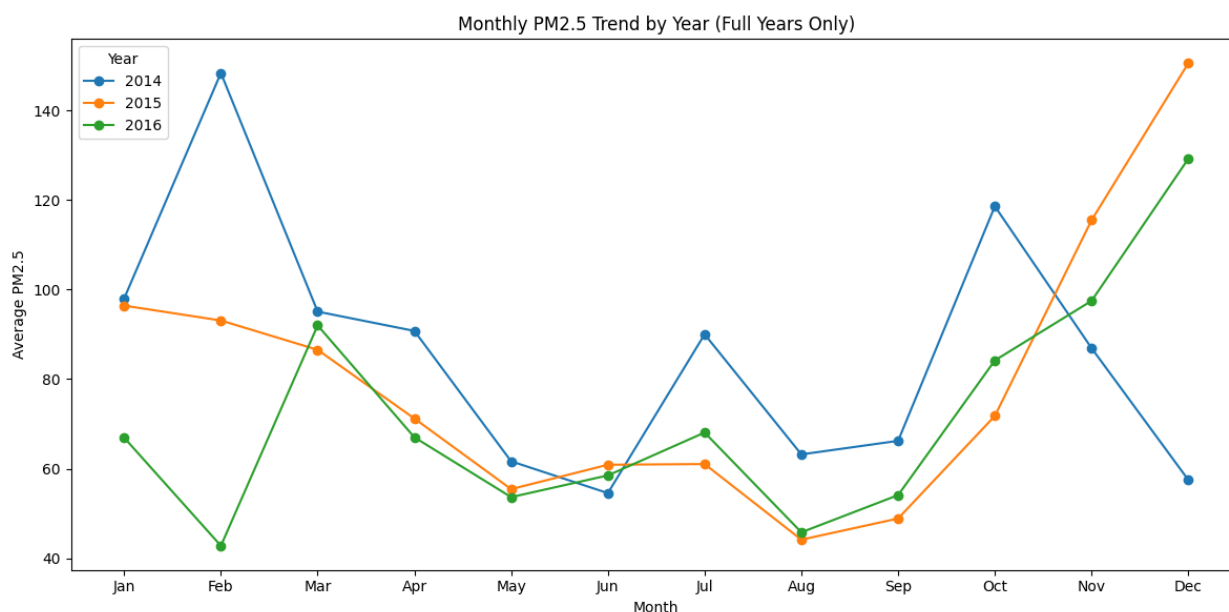
Note. Winter had the highest average PM2.5 concentration, while summer had the lowest average PM2.5 concentration.

Figure 4*PM2.5 Distribution by Season*

Note. The boxplot shows that PM2.5 varies by season and contains high outlier values in each season.

Time-Series PM2.5 Patterns

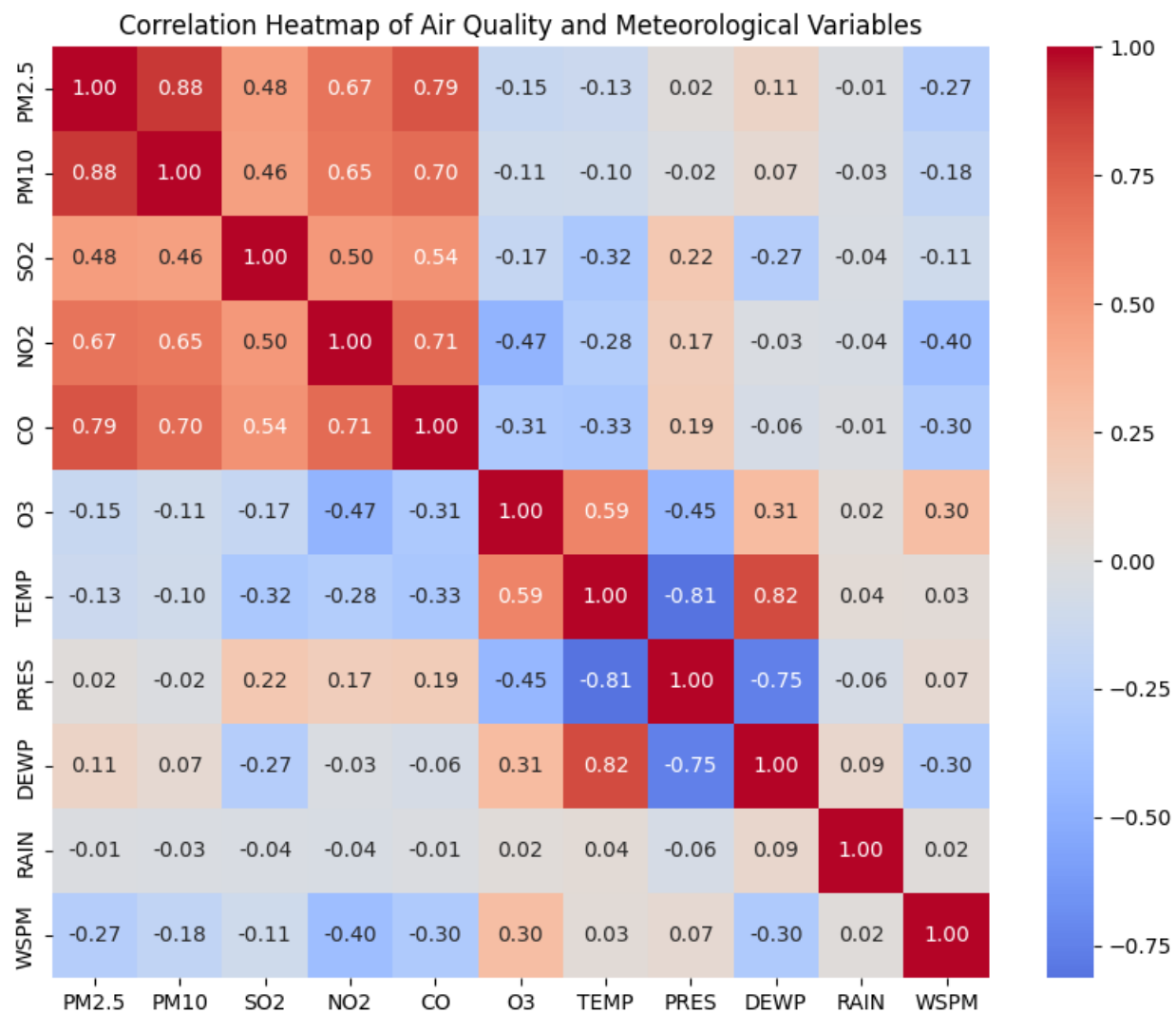
Monthly trends by year further support the presence of recurring seasonal patterns. To support fair year-over-year comparison, the notebook used complete years only: 2014, 2015, and 2016. The resulting line plot shows recurring winter peaks and summer reductions in PM2.5. The pattern appears more seasonal than year-specific, suggesting that seasonal timing is an important factor for predicting PM2.5.

Figure 5*Monthly PM2.5 Trend by Year*

Note. The chart uses full years only and shows recurring winter peaks and summer decreases across multiple years.

Correlation and Wind-Speed Relationships

Correlation analysis was used to identify variables that may be important for modeling PM2.5. PM2.5 had strong positive relationships with PM10 ($r = 0.88$), CO ($r = 0.79$), and NO2 ($r = 0.67$). Wind speed had a negative relationship with PM2.5 ($r = -0.27$). These findings suggest that pollutant variables are strongly related to each other, while stronger winds are associated with lower PM2.5 concentrations.

**Figure 6**

Correlation Heatmap of Air Quality and Meteorological Variables

Note. PM2.5 is strongly correlated with other pollutant variables and negatively correlated with wind speed.

The scatterplot of PM2.5 and wind speed shows that severe PM2.5 episodes occur primarily during low wind conditions. This pattern supports the idea that wind can help disperse pollutants and reduce the likelihood of extreme particulate matter concentration.

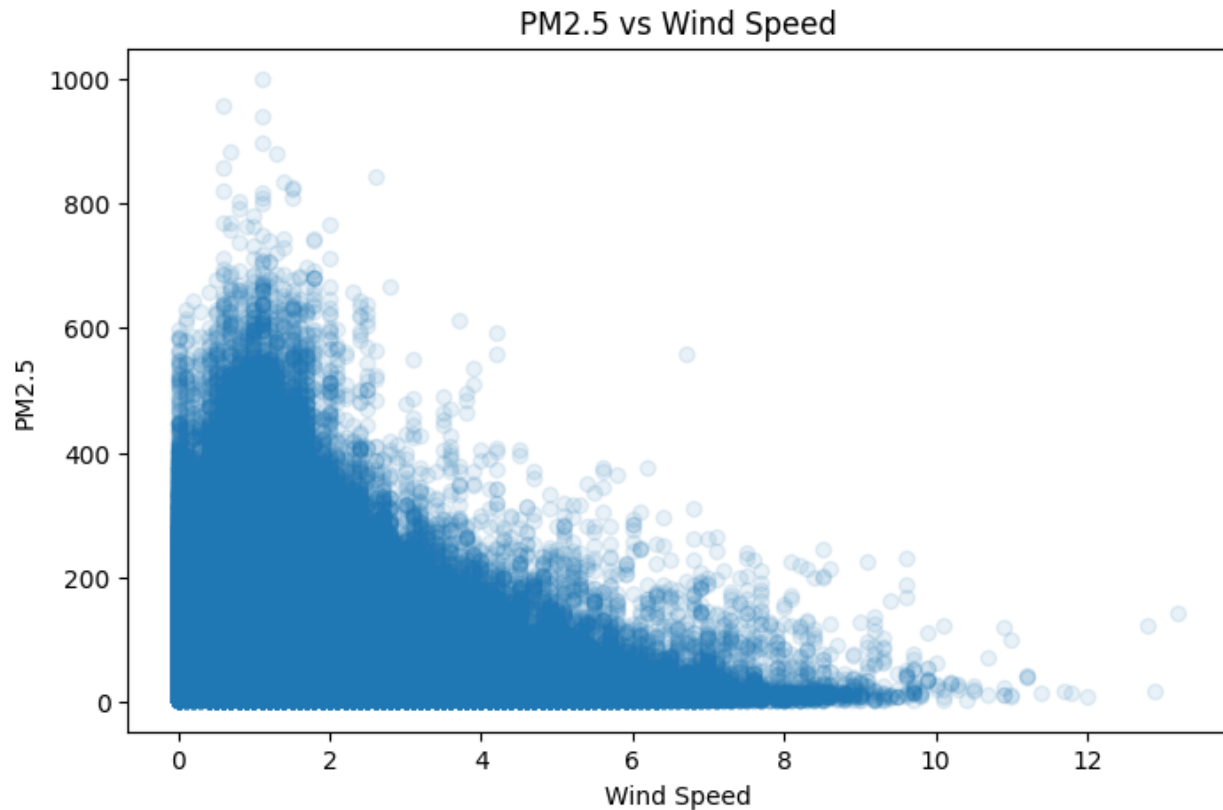


Figure 7

PM2.5 and Wind Speed Relationship

Note. Extreme PM2.5 values appear more common when wind speed is low.

Statistical Analysis

Inferential statistical tests were conducted to determine whether observed EDA patterns were statistically meaningful. These tests help connect descriptive findings to probability and statistics concepts covered in the course.

Seasonal Differences in PM2.5

The first hypothesis test examined whether PM2.5 concentrations differed significantly across seasons. Because there are four seasonal groups, a one-way analysis of variance (ANOVA) was used.

Null hypothesis: Mean PM2.5 concentration is the same across winter, spring, summer, and fall.

Alternative hypothesis: At least one season has a different mean PM2.5 concentration.

The ANOVA test produced $F = 2617.12$ and $p < .001$. Therefore, the null hypothesis was rejected. This result indicates that PM2.5 concentrations differ significantly across seasons. This supports the EDA finding that winter has the highest average PM2.5 and summer has the lowest average PM2.5.

Association Between Wind Speed and PM2.5

The second hypothesis test examined the relationship between wind speed and PM2.5 concentration. A Pearson correlation significance test was used because both variables are numeric.

Null hypothesis: Wind speed has no significant relationship with PM2.5 concentration.

Alternative hypothesis: Wind speed is significantly associated with PM2.5 concentration.

The Pearson correlation test produced $r = -0.27$ and $p < .001$. Therefore, the null hypothesis was rejected. The negative correlation indicates that higher wind speeds are generally associated with lower PM2.5 concentrations. Although the relationship is not extremely strong, it is statistically significant and directionally meaningful for air pollution interpretation.

Model Selection

The modeling objective was to predict PM2.5 concentration or PM2.5 pollution category using pollutant, meteorological, and time-related variables. PM2.5 was selected as the target variable because it is an important air-quality indicator and because the EDA demonstrated strong variation by station, season, year, and weather conditions.

Two prediction settings were evaluated. The first setting was a nowcasting task, which predicted PM2.5 using concurrent pollutant and weather readings, including SO₂, NO₂, CO, O₃, temperature, pressure, dew point, rainfall, and wind speed. The second setting was a forecasting-style task, which predicted PM2.5 using weather and time features only, without concurrent pollutant measurements. Comparing these settings is useful because nowcasting represents the best available information scenario, while the weather-only setting is closer to a future prediction problem.

For model comparison, PM10 was excluded because it was highly correlated with PM2.5 and could dominate the model. Wind direction was also excluded. Hour and month were encoded as sine and cosine features to preserve cyclical time patterns. An 80/20 temporal split was used, meaning the earlier 80% of observations formed the training set and the final 20% formed the test set. This avoided shuffling future observations into training and reduced the risk of temporal data leakage.

The model-selection notebook evaluated both regression and classification formulations. For regression, PM2.5 remained a continuous numeric target and models were evaluated using root mean squared error (RMSE), mean absolute error (MAE), and R -squared. For classification, PM2.5 was discretized into quintile classes and binary high/low classes using the total dataset distribution. Classification models were evaluated using accuracy and weighted F_1 score. The models evaluated included ordinary least squares linear regression, random forest regression/classification, logistic regression, Gaussian naive Bayes, and XGBoost. XGBoost is a gradient-boosted tree ensemble that sequentially fits trees to reduce prior model errors and can capture nonlinear relationships and interactions (T. Chen & Guestrin, 2016).

Table 5

Regression Model Performance for All Stations

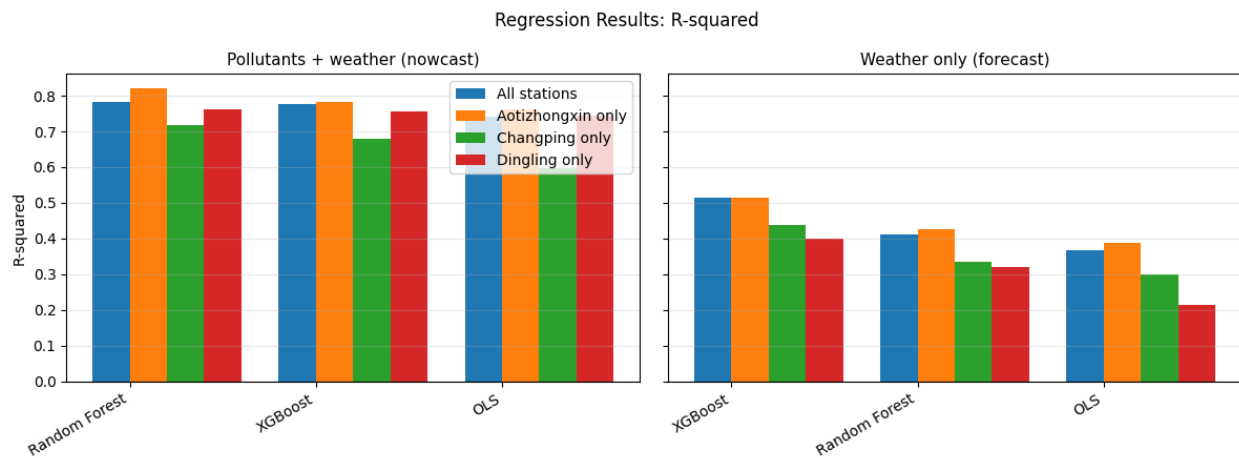
Scenario	Model	RMSE	MAE	R^2
Nowcast: pollutants + weather	XGBoost	38.945	23.395	0.778
Nowcast: pollutants + weather	Random Forest	38.666	23.426	0.781
Nowcast: pollutants + weather	OLS	42.079	27.956	0.741
Forecast: weather only	XGBoost	57.558	39.889	0.515
Forecast: weather only	Random Forest	63.416	43.562	0.411
Forecast: weather only	OLS	65.754	46.713	0.367

Note. Lower RMSE and MAE indicate lower prediction error. Higher R^2 indicates a larger proportion of PM2.5 variance explained.

For the all-station regression task, XGBoost was best or comparable. In the nowcast setting, XGBoost achieved $R^2 = 0.778$, RMSE = 38.945, and MAE = 23.395. In the weather-only forecast setting, performance declined to $R^2 = 0.515$, RMSE = 57.558, and MAE = 39.889. This decline is expected because the forecast setting removes concurrent pollutant measurements that provide strong direct signals about air quality.

Figure 8

Regression Results: R-Squared Comparison



Note. XGBoost produced the highest R -squared in both the nowcast and weather-only forecast scenarios.

Table 6

XGBoost Classification Performance for All Stations

Scenario	Classification task	Accuracy	Weighted F_1
Nowcast: pollutants + weather	PM2.5 quintile	0.616	0.613
Nowcast: pollutants + weather	PM2.5 binary	0.870	0.870
Forecast: weather only	PM2.5 quintile	0.456	0.434
Forecast: weather only	PM2.5 binary	0.768	0.768

Note. Binary classification separates PM2.5 above and below the total dataset median.

Quintile classification separates PM2.5 into five ordered groups.

Classification results followed the same overall pattern as regression. Binary classification was much easier than quintile classification because it only required the model to separate higher and lower PM2.5 observations. XGBoost achieved 0.870 accuracy and 0.870 weighted F_1 in the nowcast binary task and 0.768 accuracy and 0.768 weighted F_1 in the weather-only binary task. Quintile classification was more difficult, especially in the middle bins where PM2.5 values are less clearly separated.

The model-selection analysis also compared pooled all-station models against single-station models for each monitoring site. XGBoost was strictly superior in all forecasting tasks and outperformed in a majority of nowcasting tasks as well. Notably, for certain highly linear stations, OLS provided comparable and sometimes superior results, showing the weakness of tree models for highly linear datasets and those with outlier data which OLS can effectively extrapolate. For some monitoring sites, Random Forest outperformed XGBoost, likely due to the tendency of XGBoost to overfit to noise for simpler datasets. In most cases, the pooled model performed similarly or better than the corresponding single-station model, indicating that a general model across sites is viable. For instance, for the Aotizhongxin station, the pooled model achieved an R^2 of 0.818 in the nowcast setting, while the single-station model achieved 0.820; however, for other stations, the pooled model often yielded higher R^2 values. One station-level result is provided in the appendix and comprehensive station-level analysis is included in the notebook.

Model Analysis

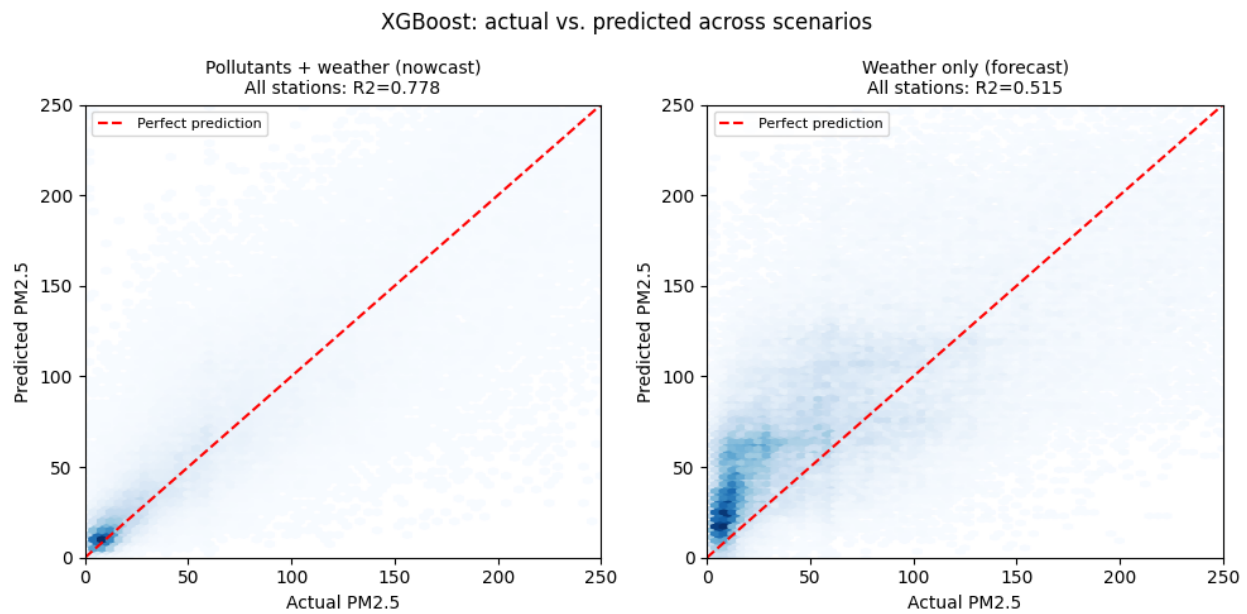
XGBoost was selected as the final model family because it showed the strongest and most consistent performance across regression, quintile classification, and binary classification tasks. It also performed well in both the nowcast and forecast-style settings. This consistency is important because a strong final model should not only perform well under one narrow scenario; it should also remain competitive when features are limited or when the target is reframed as a classification problem.

The model results show a clear distinction between nowcasting and forecasting. The nowcast model performed best because concurrent pollutant measurements such as CO, NO2,

SO₂, and O₃ provide strong direct information about current PM_{2.5} conditions. The weather-only model performed worse but still explained approximately half of the PM_{2.5} variation in the all-station regression task. This suggests that meteorological and seasonal conditions are meaningful predictors, but weather alone cannot fully identify emission events or short-term pollution spikes.

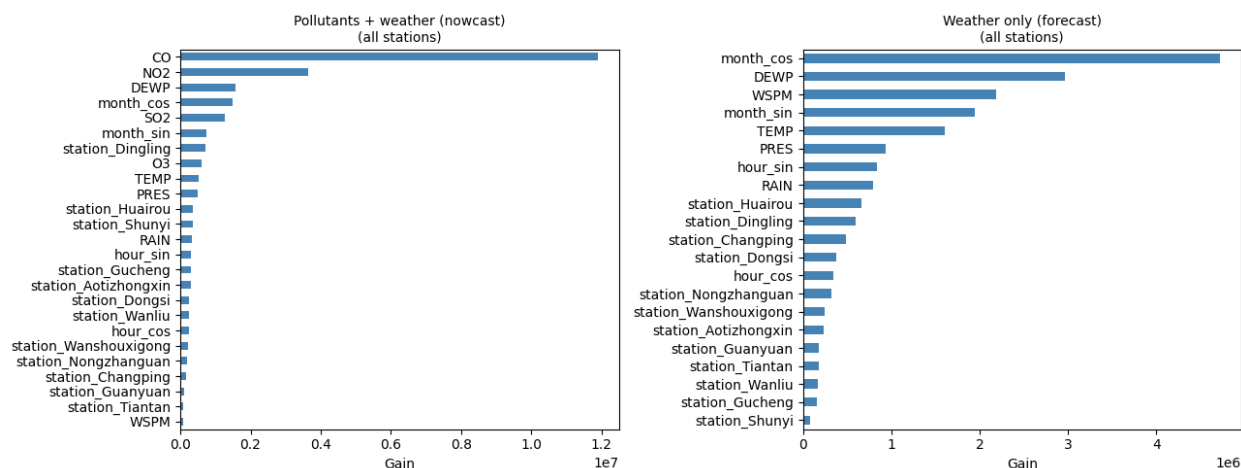
Figure 9

XGBoost Actual Versus Predicted PM_{2.5}



Note. Predictions closer to the diagonal line indicate better model fit. Nowcast scenarios show stronger alignment than weather-only forecast scenarios.

Feature-importance output further supports the EDA findings. In the nowcast scenario, pollutant features, especially CO and NO₂, contributed strongly to model performance. In the weather-only forecast scenario, temporal features, dew point, pressure, and other meteorological variables became more important. This suggests that the nowcast model relies on direct pollutant signals, while the forecast-style model relies more heavily on seasonal and weather patterns.

Figure 10*XGBoost Feature Importance*

Note. The nowcast model is driven more by pollutant variables, while the weather-only model relies more on time and meteorological variables.

The selected model is valid for the course project because it is evaluated using a temporal train/test split, compared against simpler baseline models, and interpreted using multiple metrics. However, there are important limitations. First, the nowcast model should not be interpreted as a true future forecast because it uses concurrent pollutant measurements. Second, the weather-only forecast model performs less strongly and may miss sudden emission events. Third, PM10 and lagged PM2.5 were excluded to prevent them from dominating the model comparison, but these variables could improve predictive performance in a real operational system. Fourth, the model does not include external predictors such as traffic volume, industrial activity, policy changes, holidays, or regional weather systems.

Overall, XGBoost provides the best balance of predictive performance and flexibility for this dataset. It can capture nonlinear relationships, tolerate correlated predictors better than linear models, and provide feature-importance outputs that support interpretation. The most appropriate final conclusion is that XGBoost is the strongest model family for PM2.5 prediction in this project, with the nowcast version performing best when concurrent pollutant readings are available and the

weather-only version offering a more realistic but less precise forecasting baseline.

Conclusion and Recommendations

This project analyzed the Beijing Multi-Site Air Quality dataset to understand PM2.5 pollution patterns and develop a predictive modeling workflow. The data cleaning process combined 12 station-level files, created a datetime feature, engineered seasonal groupings, and treated missing values using station-level median and mode imputation. The cleaned dataset supported exploratory analysis, statistical testing, and model development.

The EDA showed that PM2.5 is right-skewed, varies across monitoring stations, is highest in winter and lowest in summer, and has recurring seasonal patterns across years. Correlation analysis showed strong positive relationships between PM2.5 and PM10, CO, and NO2, while wind speed had a significant negative relationship with PM2.5. Statistical testing confirmed that PM2.5 differs significantly across seasons and that wind speed is significantly associated with PM2.5.

The model-selection results support XGBoost as the final model family. In the all-station nowcast regression task, XGBoost achieved $R^2 = 0.778$, RMSE = 38.945, and MAE = 23.395. In the weather-only forecast-style task, XGBoost remained the best model but declined to $R^2 = 0.515$. This gap is meaningful: it shows that concurrent pollutant readings substantially improve current PM2.5 estimation, while weather and time variables provide useful but incomplete predictive information.

From a practical perspective, this type of analysis can support environmental monitoring, public health planning, and early warning systems for poor air quality. A nowcast model could support real-time air-quality estimation when multiple pollutant measurements are available. A weather-only model could serve as a baseline for forecasting periods when pollutant data are unavailable or delayed. Future work should add lagged PM2.5, traffic activity, industrial activity, policy changes, holidays, and additional regional weather measures. Future model evaluation should also focus more specifically on high-pollution tail events, because correctly identifying severe pollution episodes may be more important for public health decision-making than

minimizing average prediction error.

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Appendix A

Technical Notebook Output Summary

The appendix summarizes key outputs from the project notebooks. The full notebooks are maintained in the team GitHub repository. The report body includes the main figures and tables needed to support the analysis.

Table A1

Key Technical Outputs Included in the Current Draft

Notebook area	Current output included
Data loading	12 station CSV files loaded and combined
Combined dataset	420,768 observations; 12 stations; March 2013–February 2017
Missing values	Missing count and percentage by variable
Feature engineering	datetime, season, hour_sin/hour_cos, and month_sin/month_cos columns created
EDA figures	PM2.5 distribution, station chart, seasonal charts, correlation heatmap, wind-speed scatterplot, monthly trend
Statistical tests	ANOVA for seasonal PM2.5 differences; Pearson correlation for wind speed and PM2.5
Modeling	Regression, quintile classification, and binary classification model comparisons

Note. Outputs are summarized from the final-modeling notebook.

Appendix B

Complete Model Comparison Output

Table B1*Regression Performance Across Model Scenarios*

Feature set		Scope	Model	RMSE	MAE	R^2
Pollutants	+	All stations	Random Forest	38.666	23.426	0.781
weather						
Pollutants	+	All stations	XGBoost	38.945	23.395	0.778
weather						
Pollutants	+	All stations	OLS	42.079	27.956	0.741
weather						
Pollutants	+	Aotizhongxin	XGBoost	35.390	22.274	0.820
weather						
Pollutants	+	Aotizhongxin	OLS	38.857	26.141	0.783
weather						
Pollutants	+	Aotizhongxin	Random Forest	40.723	24.799	0.762
weather						
Weather only		All stations	XGBoost	57.558	39.889	0.515
Weather only		All stations	Random Forest	63.416	43.562	0.411
Weather only		All stations	OLS	65.754	46.713	0.367
Weather only		Aotizhongxin	XGBoost	58.036	39.994	0.516
Weather only		Aotizhongxin	Random Forest	63.200	43.952	0.426
Weather only		Aotizhongxin	OLS	65.320	46.220	0.387

Note. Final-modeling notebook excerpt; feature set and scope combinations used an 80/20 temporal train/test split. Other station values included in the notebook.

Table B2*Classification Performance Across All-Stations Model Scenarios*

Feature set	Task	Model	Accuracy	Weighted F_1
Pollutants + weather	Quintile	XGBoost	0.616	0.613
Pollutants + weather	Quintile	Logistic Regression	0.602	0.601
Pollutants + weather	Quintile	Random Forest	0.567	0.556
Pollutants + weather	Quintile	Naive Bayes	0.483	0.462
Pollutants + weather	Binary	XGBoost	0.870	0.870
Pollutants + weather	Binary	Logistic Regression	0.857	0.856
Pollutants + weather	Binary	Random Forest	0.845	0.845
Pollutants + weather	Binary	Naive Bayes	0.770	0.764
Weather only	Quintile	XGBoost	0.456	0.434
Weather only	Quintile	Logistic Regression	0.443	0.425
Weather only	Quintile	Random Forest	0.395	0.372
Weather only	Quintile	Naive Bayes	0.341	0.320
Weather only	Binary	XGBoost	0.768	0.768
Weather only	Binary	Logistic Regression	0.729	0.728
Weather only	Binary	Random Forest	0.632	0.617
Weather only	Binary	Naive Bayes	0.615	0.613

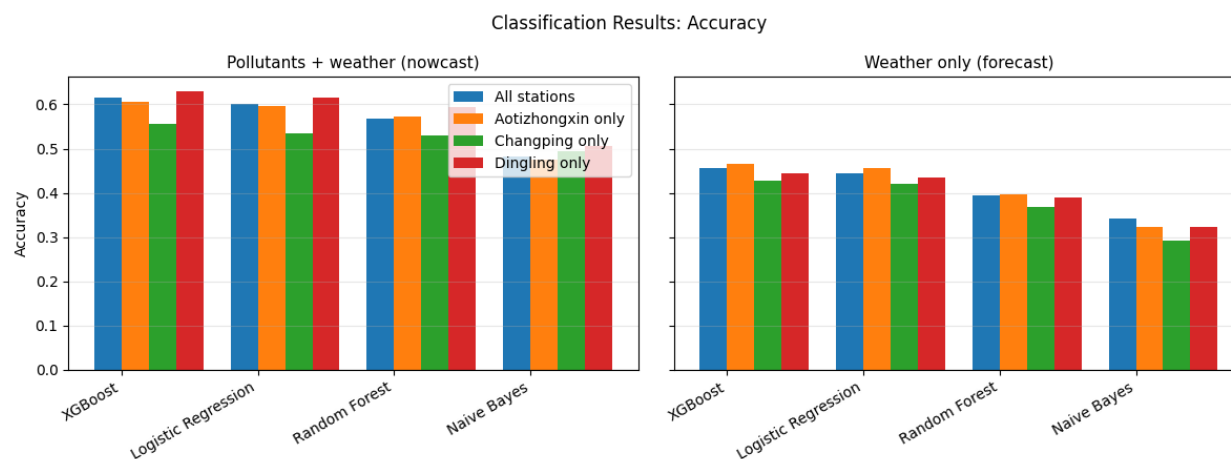
Note. Classification results shown for pooled all-station experiments only. The binary task separates PM_{2.5} into below/above median categories based on complete dataset.

Appendix C

Additional Model Figures

Figure C1

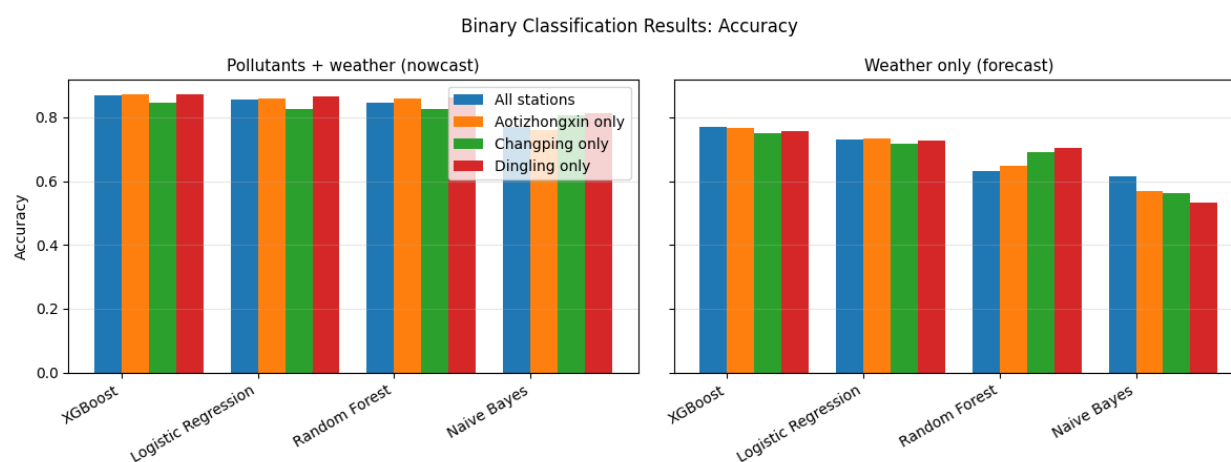
Quintile Classification Results: Accuracy



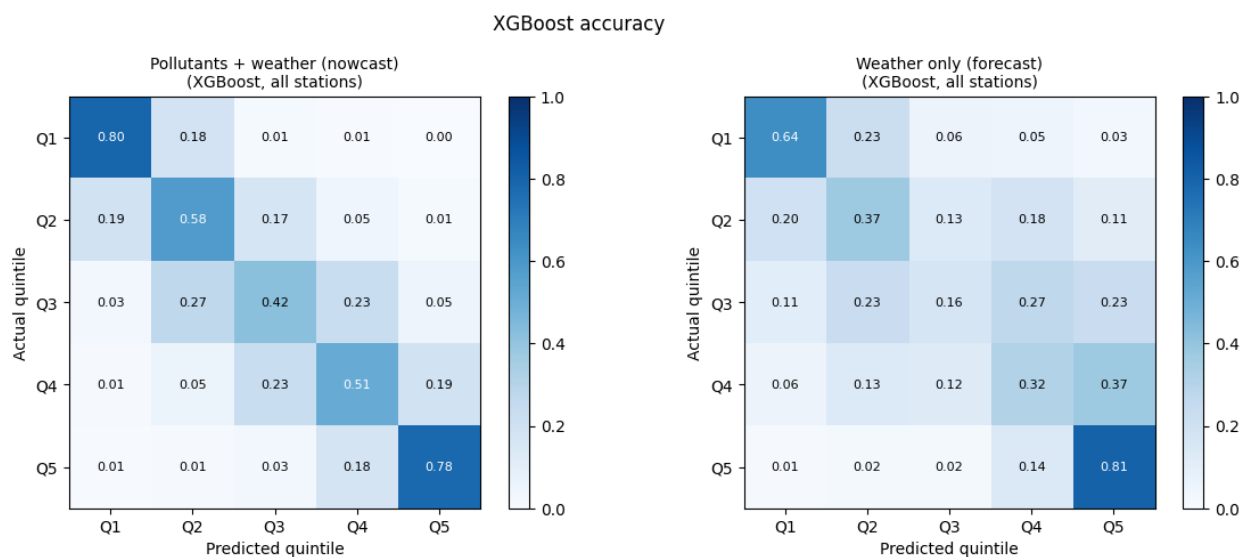
Note. Quintile classification is harder than binary classification, particularly in the weather-only forecast setting.

Figure C2

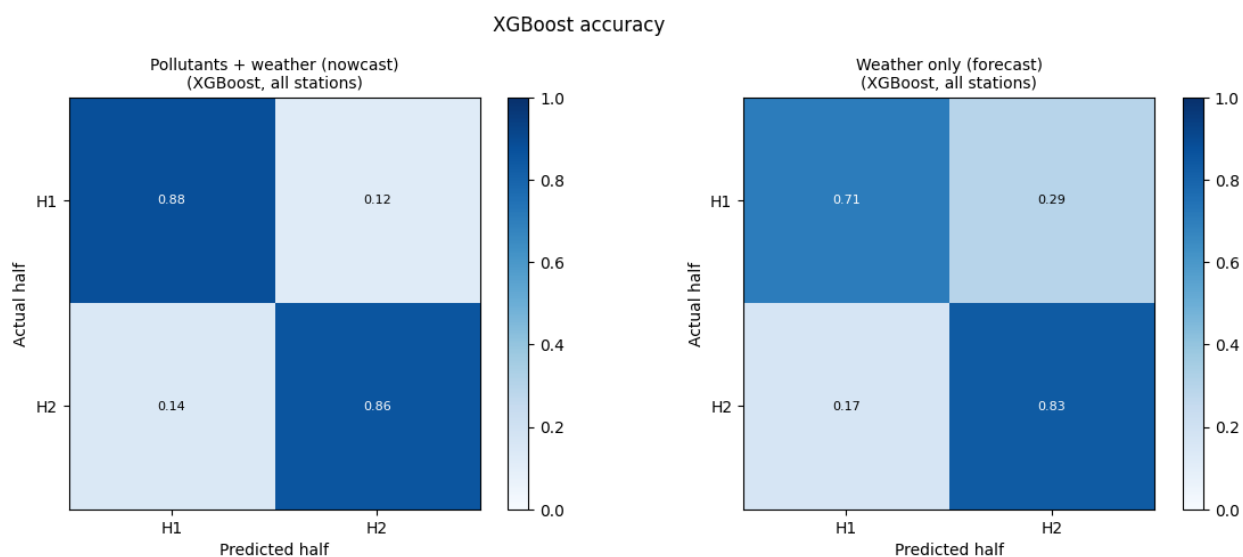
Binary Classification Results: Accuracy



Note. Binary classification accuracy is substantially above the 50% chance baseline for all model scenarios.

Figure C3*XGBoost Quintile Classification Confusion Matrix*

Note. Extreme quintiles are generally easier to classify than middle quintiles.

Figure C4*XGBoost Binary Classification Confusion Matrix*

Note. The binary confusion matrix shows stronger class separation than the five-quintile task.

Appendix D

Code and Repository Information

GitHub repository: <https://github.com/sudhakaran-srinivasan/air-quality-statistical-modeling>

The relevant notebook in the repository is `final_modeling.ipynb`.